

Supplementary Materials

Tables:

Table S1. Image features extracted from segmented needle litter regions and their interpretation

ID	Feature group	Feature	Feature description	Interpretation / relevance
1	Geometric Features	area	Number of pixels classified as needle litter within the segmented mask	Represents the projected cover area of needle litter; generally related to litter coverage and loading
2		perimeter	Boundary length of the segmented needle litter region	Reflects the edge complexity and spatial discontinuity of the needle litter region
3		compactness	Shape compactness calculated from area and perimeter	Describes the aggregation or fragmentation degree of the needle litter region
4		RGB_R_mean	Mean value of the red, green, and blue channel within the needle region	Describes the average red tone of needle litter
5		RGB_G_mean		Describes the average green tone and possible influence of background vegetation or fresh material
6		RGB_B_mean		Describes the average blue component of the needle region
7		RGB_R_var	Variance of the red, green, and blue channel within the needle region	Reflects color heterogeneity in red, green, and blue channel
8	Color Features	RGB_G_var		
9		RGB_B_var		
10		RGB_R_skew	Skewness of the red, green, and blue channel distribution	Describes asymmetry of red-channel, green-channel, and blue-channel pixel values
11		RGB_G_skew		
12		RGB_B_skew		
13		LAB_L_mean	Mean value of the L, A, and B channel in LAB color space	Represents average brightness of the needle region
14		LAB_A_mean		Describes green–red and blue–yellow colors component of needle litter
15		LAB_B_mean		Reflects brightness heterogeneity caused by illumination, shadow and litter overlap
16		LAB_L_var	Variance of the L, A, and B channel	Reflects heterogeneity of green–red and blue–yellow colors information
17		LAB_A_var		
18		LAB_B_var		
19	Texture Features	LAB_L_skew	Skewness of the L, A, and B channel	Describes asymmetry of brightness, green–red color, and blue–yellow color distribution.
20		LAB_A_skew		
21		LAB_B_skew		
22		LBP_bin0	Histogram proportion of local binary pattern bin 0-9	Represents one type of local micro-texture pattern in the needle region
23		LBP_bin1		
24		LBP_bin2		
25		LBP_bin3		
26		LBP_bin4		
27		LBP_bin5		
28		LBP_bin6		
29		LBP_bin7		
30		LBP_bin8		
31		LBP_bin9		
32		GLCM_contrast	Gray-level co-occurrence matrix contrast, energy, and correlation.	Reflects local gray-level variation and textural roughness
33		GLCM_energy		Reflects textural uniformity and gray-level concentration
34		GLCM_correlation		Reflects spatial dependence among neighboring gray levels

Note: Geometric features were calculated from the binary mask of segmented needle litter. Color features were extracted from pixels within the segmented needle region in both RGB and LAB color spaces. Mean, variance and skewness were used to describe the central tendency, heterogeneity and distributional asymmetry of color values, respectively. Texture features included a 10-bin local binary pattern histogram and three gray-level co-occurrence matrix descriptors, which were used to characterize the local texture, roughness and spatial arrangement of needle litter.

Table S2. Variable importance of the 34 image features in the indoor RF source model.

ID	Feature	Overall	ID	Feature	Overall	ID	Feature	Overall
1	area	71.057	13	LAB_L_mean	69.838	25	LBP_bin0	19.927
2	perimeter	58.500	14	LAB_L_var	60.498	26	LBP_bin1	49.783
3	compactness	71.989	15	LAB_L_skew	59.707	27	LBP_bin2	32.508
4	RGB_R_mean	96.436	16	LAB_A_mean	45.396	28	LBP_bin3	56.427
5	RGB_R_var	66.242	17	LAB_A_var	24.344	29	LBP_bin4	77.910
6	RGB_R_skew	82.799	18	LAB_A_skew	22.622	30	LBP_bin5	42.112
7	RGB_G_mean	83.538	19	LAB_B_mean	39.496	31	LBP_bin6	0.000
8	RGB_G_var	79.562	20	LAB_B_var	100.000	32	LBP_bin7	21.013
9	RGB_G_skew	72.419	21	LAB_B_skew	22.347	33	LBP_bin8	35.384
10	RGB_B_mean	26.894	22	GLCM_contrast	44.051	34	LBP_bin9	44.196
11	RGB_B_var	64.676	23	GLCM_energy	61.958			
12	RGB_B_skew	54.955	24	GLCM_correlation	53.158			

Table S3. Indoor–field feature shifts and changes in feature–load relationships.

(A) Feature-distribution shifts between indoor and field scenes.

Feature	Indoor mean $\pm$ SD	Field mean $\pm$ SD	Std. Shift	Cliff's δ	P_value	FDR-adjusted P
LAB_A_var	3.580 $\pm$ 0.806	7.724 $\pm$ 2.275	5.141	0.951	0.000	0.000
LAB_B_skew	0.0885 $\pm$ 0.0925	0.4165 $\pm$ 0.3365	3.545	0.827	0.000	0.000
LBP_bin2	0.0485 $\pm$ 0.0018	0.0428 $\pm$ 0.0033	-3.226	-0.862	0.000	0.000
LAB_A_mean	133.75 $\pm$ 0.966	136.41 $\pm$ 2.971	2.745	0.536	0.000	0.000
RGB_R_mean	172.97 $\pm$ 3.185	181.39 $\pm$ 5.034	2.644	0.885	0.000	0.000
LBP_bin9	0.0870 $\pm$ 0.0066	0.1001 $\pm$ 0.0106	1.976	0.701	0.000	0.000
RGB_B_mean	133.72 $\pm$ 4.193	140.65 $\pm$ 6.123	1.652	0.653	0.000	0.000
LBP_bin5	0.1284 $\pm$ 0.0078	0.1156 $\pm$ 0.0145	-1.635	-0.537	0.000	0.000
LAB_B_mean	140.11 $\pm$ 0.922	138.87 $\pm$ 1.615	-1.337	-0.456	0.000	0.000
LBP_bin8	0.0130 $\pm$ 0.0019	0.0156 $\pm$ 0.0030	1.336	0.522	0.000	0.000
LBP_bin0	0.0669 $\pm$ 0.0075	0.0767 $\pm$ 0.0094	1.31	0.568	0.000	0.000
RGB_R_skew	-0.1583 $\pm$ 0.0745	-0.2549 $\pm$ 0.0640	-1.296	-0.654	0.000	0.000
LBP_bin7	0.0240 $\pm$ 0.0033	0.0277 $\pm$ 0.0050	1.109	0.448	0.000	0.000
LAB_L_mean	161.35 $\pm$ 4.329	166.06 $\pm$ 4.189	1.088	0.526	0.000	0.000
GLCM_contrast	10.308 $\pm$ 1.370	11.795 $\pm$ 1.857	1.085	0.442	0.000	0.000
RGB_R_var	2686.2 $\pm$ 177.27	2494.2 $\pm$ 162.33	-1.083	-0.590	0.000	0.000
LBP_bin1	0.0944 $\pm$ 0.0059	0.1006 $\pm$ 0.0102	1.046	0.36	0.000	0.000
LAB_B_var	6.612 $\pm$ 1.934	8.547 $\pm$ 1.875	1.001	0.512	0.000	0.000
LAB_L_var	2566.0 $\pm$ 184.21	2396.5 $\pm$ 165.36	-0.920	-0.518	0.000	0.000
GLCM_correlation	0.8022 $\pm$ 0.0195	0.7850 $\pm$ 0.0320	-0.882	-0.310	0.000	0.000
RGB_G_var	2717.0 $\pm$ 183.90	2586.6 $\pm$ 158.82	-0.709	-0.432	0.000	0.000
LBP_bin3	0.1361 $\pm$ 0.0066	0.1314 $\pm$ 0.0101	-0.709	-0.278	0.000	0.000
RGB_G_mean	150.62 $\pm$ 4.870	153.74 $\pm$ 5.444	0.641	0.291	0.000	0.000
LBP_bin4	0.3600 $\pm$ 0.0191	0.3485 $\pm$ 0.0173	-0.602	-0.343	0.000	0.000
LAB_L_skew	-0.2242 $\pm$ 0.0877	-0.2726 $\pm$ 0.0804	-0.552	-0.281	0.000	0.000
RGB_G_skew	-0.1166 $\pm$ 0.0874	-0.1541 $\pm$ 0.0839	-0.430	-0.218	0.004	0.005
RGB_B_var	2556.3 $\pm$ 184.69	2487.9 $\pm$ 179.71	-0.371	-0.220	0.003	0.004
area	1109538.7 $\pm$ 107152.6	1071206.8 $\pm$ 80606.7	-0.358	-0.197	0.008	0.010
compactness	0.0240 $\pm$ 0.0183	0.0176 $\pm$ 0.0074	-0.347	0.008	0.910	0.910
GLCM_energy	0.2841 $\pm$ 0.0575	0.3040 $\pm$ 0.0451	0.346	0.203	0.007	0.008
RGB_B_skew	-0.1088 $\pm$ 0.0893	-0.1360 $\pm$ 0.0861	-0.304	-0.162	0.030	0.034
LAB_A_skew	-0.1098 $\pm$ 0.2360	-0.1748 $\pm$ 0.3642	-0.275	-0.069	0.359	0.381
LBP_bin6	0.0416 $\pm$ 0.0043	0.0409 $\pm$ 0.0046	-0.170	-0.091	0.221	0.243
perimeter	29497.8 $\pm$ 9557.6	29248.5 $\pm$ 5131.6	-0.026	-0.028	0.707	0.728

(B) Changes in Spearman correlations between image features and needle litter load.

Feature	Indoor ρ	Indoor FDR	Field ρ	Field FDR	$\Delta\rho$
RGB_R_mean	-0.647	0.000	0.407	0.000	1.054
RGB_B_var	0.637	0.000	-0.288	0.002	0.925
LBP_bin5	0.417	0.000	-0.488	0.000	0.905
LBP_bin1	-0.500	0.000	0.399	0.000	0.899
LAB_B_var	-0.522	0.000	0.375	0.000	0.897
LBP_bin0	-0.465	0.000	0.390	0.000	0.855
LAB_B_mean	-0.354	0.000	0.464	0.000	0.818
GLCM_contrast	-0.453	0.000	0.356	0.000	0.809
LAB_L_var	0.581	0.000	-0.199	0.044	0.780
GLCM_correlation	0.277	0.003	-0.484	0.000	0.761
LBP_bin9	-0.273	0.003	0.482	0.000	0.755
RGB_G_var	0.567	0.000	-0.185	0.057	0.752
RGB_R_var	0.576	0.000	-0.145	0.134	0.721
LBP_bin4	0.37	0.000	-0.341	0.000	0.711
LBP_bin3	-0.641	0.000	0.039	0.695	0.680
LAB_A_var	-0.110	0.239	0.532	0.000	0.642
RGB_R_skew	0.597	0.000	-0.014	0.880	0.611
LBP_bin6	0.355	0.000	-0.191	0.050	0.546
compactness	0.677	0.000	0.183	0.057	0.493
perimeter	-0.656	0.000	-0.166	0.086	0.490
LAB_L_mean	-0.603	0.000	-0.195	0.047	0.408
area	0.577	0.000	0.204	0.040	0.373
LAB_B_skew	0.106	0.248	-0.237	0.015	0.343
GLCM_energy	-0.574	0.000	-0.316	0.001	0.257
LAB_L_skew	0.569	0.000	0.349	0.000	0.221
RGB_G_mean	-0.594	0.000	-0.388	0.000	0.206
LBP_bin2	-0.324	0.000	-0.124	0.194	0.200
RGB_B_mean	-0.497	0.000	-0.336	0.000	0.161
RGB_G_skew	0.568	0.000	0.414	0.000	0.155
LAB_A_skew	-0.420	0.000	-0.571	0.000	0.151
LBP_bin8	0.211	0.022	0.091	0.343	0.120
LBP_bin7	0.252	0.006	0.141	0.141	0.110
RGB_B_skew	0.534	0.000	0.428	0.000	0.106
LAB_A_mean	0.562	0.000	0.597	0.000	0.035

Table S4. Robustness diagnostics of the MLR- and RF-based bridge-calibration models.

Analysis	Diagnostic	MLR-based bridge	RF-based bridge
Source-field association	Calibration Pearson r	0.441 ($P = 0.007$)	0.246 ($P = 0.147$)
	Calibration Spearman ρ	0.512 ($P = 0.001$)	0.222 ($P = 0.193$)
	Validation Pearson r	0.425 ($P < 0.001$)	0.049 ($P = 0.661$)
	Validation Spearman ρ	0.452 ($P < 0.001$)	-0.044 ($P = 0.694$)
Bridge-slope	Slope (95% CI)	0.059 (0.017–0.100)	0.074 (-0.027–0.175)
stability	Bootstrap 2.5th–97.5th percentile	0.026–0.108	-0.007–0.147
Prediction	SD ratio to observed	41.3%	23.6%
shrinkage	Range ratio to observed	42.5%	23.6%
Mean-baseline comparison	Median absolute error: linear bridge (t/ha)	0.458	0.482
	Median absolute error: mean baseline (t/ha)	0.508	0.508
	FDR-adjusted P	0.035	0.238
GAM complexity	Smooth-term edf	1	1.71

Figures:

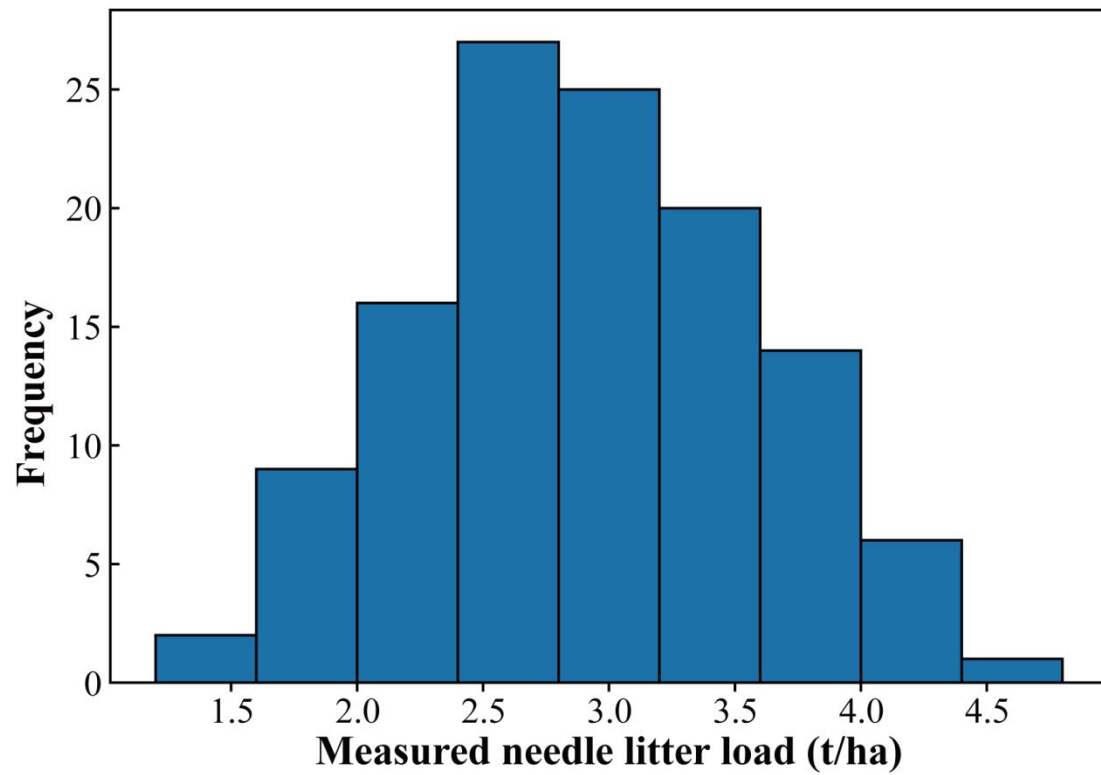


Fig. S1. Distribution of measured *P. massoniana* needle litter loads for the 120 field samples used in the cross-scene analyses.

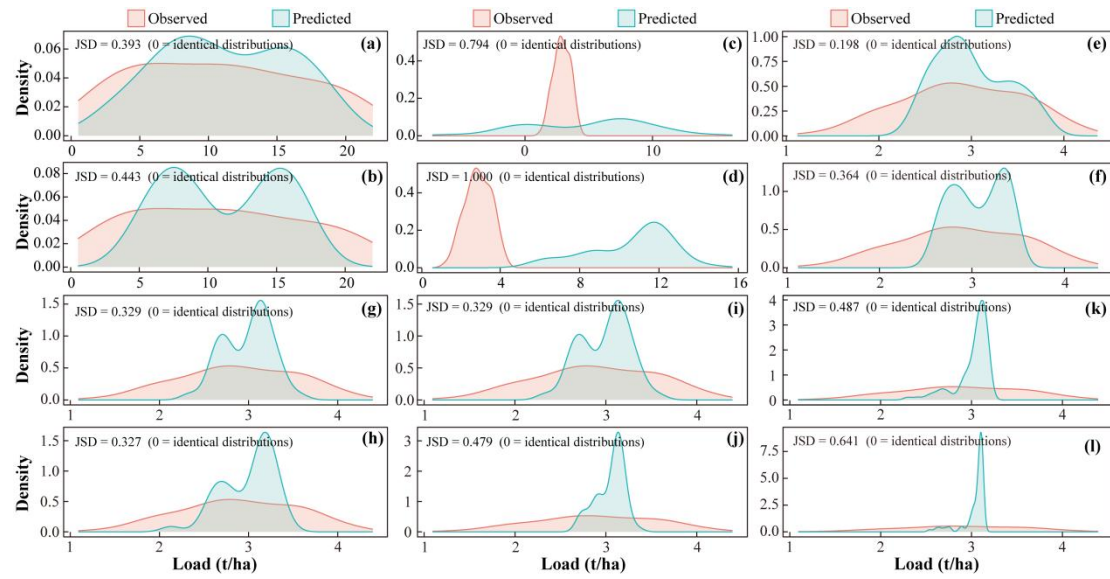


Fig. S2. Observed–predicted distribution comparisons for all regression models and field prediction strategies. (a–b) Indoor source models: (a) MLR and (b) RF; (c–d) direct extrapolation: (c) MLR and (d) RF; (e–f) field-local models: (e) MLR and (f) RF; (g–i) MLR-based bridge calibration: (g) linear, (h) quadratic, and (i) GAM; and (j–l) RF-based bridge calibration: (j) linear, (k) quadratic, and (l) GAM. JSD values are shown in each panel, with lower values indicating greater similarity between the observed and predicted distributions.

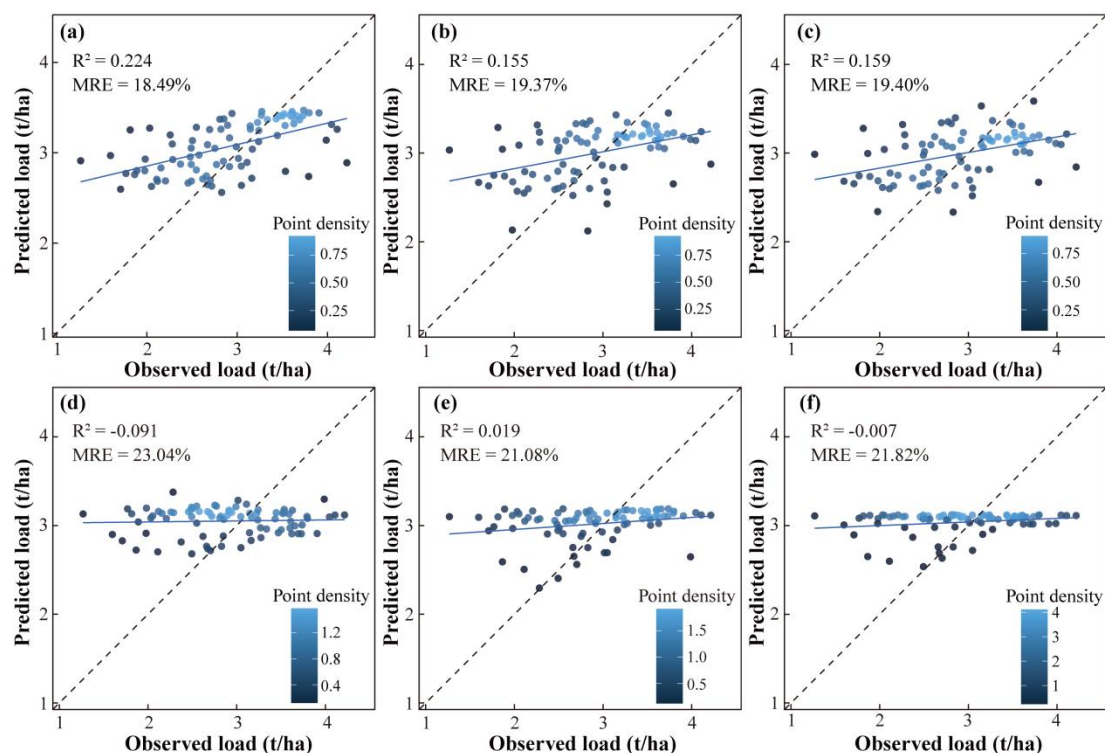


Fig. S3. Observed versus predicted needle litter loads on the common independent field validation set for the remaining field prediction models: (a) field-local RF; (b) MLR-based quadratic bridge; (c) MLR-based GAM bridge; (d) RF-based linear bridge; (e) RF-based quadratic bridge; and (f) RF-based GAM bridge. Plot conventions are as in Fig. 4.